Solution

Given that $P(A) = 0.6, \quad P(B) = 0.5, \quad$ and $\quad P(A \cap B) = 0.2$

(i) $$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = \frac{2}{5}$$

(ii) $$P(\overline{A}/B) = \frac{P(\overline{A} \cap B)}{P(B)}$$

$$= \frac{P(B) - P(A \cap B)}{P(B)}$$

$$= \frac{0.5 - 0.2}{0.5} = \frac{0.3}{0.5} = \frac{3}{5}.$$

(iii) $$P(A/\overline{B}) = \frac{P(A \cap \overline{B})}{P(\overline{B})}$$

$$= \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

$$= \frac{0.6 - 0.2}{1 - 0.5} = \frac{0.4}{0.5} = \frac{4}{5}.$$

Note 12.5

$$P(A/B) + P(\overline{A}/B) = 1$$

Example 12.17

A die is rolled. If it shows an odd number, then find the probability of getting 5.

Solution

Sample space $S = \{1, 2, 3, 4, 5, 6\}$.

Let A be the event of die shows an odd number.

Let B be the event of getting 5.

Then, $A = \{1, 3, 5\}$, $B = \{5\}$, and $A \cap B = \{5\}$.

Therefore, $P(A) = \frac{3}{6}$ and $P(A \cap B) = \frac{1}{6}$

$P(\text{getting } 5 / \text{die shows an odd number}) = P(B/A)$

$$= \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{6}}{\frac{3}{6}}$$

$$P(B/A) = \frac{1}{3}.$$

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